

Sep 2, 2025
COL 7341/774

Name Buaya:-

$p(x, y)$

$x = (x_1, \dots, x_n)$
 $\hookrightarrow$ descort can $\left[\begin{matrix} x_{j1} \\ x_{j2} \\ \vdots \\ x_{jn} \end{matrix} \right]$ x_j

$y \in \{0, 1\}$

$f(x)$ $y \in \{0, 1\}$

$\mathbb{P} \rightarrow x_1, \dots, x_n | y$

[Attributes are independent of each other given class]

$y \sim \text{Bernoulli}(\phi)$
 $x | y \sim N(\mu_y, \Sigma_y)$
 $y \in \{0, 1\}$

$n \times n \begin{bmatrix} \sigma^2 \end{bmatrix}$

Gaussian Naive Bayes:-

$$P(y) = \prod_{i=1}^n P(x_i | y) = P(x_1 | y) \dots P(x_n | y)$$

$$y \sim \text{Bernoulli}(\theta) \quad x_i | y \sim N(\mu_y, \sigma_y^2)$$

$$P(x_i | y) = \frac{1}{\sigma_y \sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right)$$

diag of Σ_y

⇒ Discrete / Cont set of attributes

Discrete

$y \in \{U, F, T\}$

$P(y|x)$

Cough

Cold

Fever

Body Ache

↓
{0,1}

↓
{0,1}

↓
{0,1}

↓
{0,1}

Naive Bayes Assumption

→ $x_i \perp x_j | y$

⇒

$$y \sim \text{Multinoulli}(\Phi) \rightarrow$$

$$x_i | y \sim \text{Multinoulli}(\theta_{yi})$$

$$P(y|x) \propto \frac{P(x|y) P(y)}{P(x)} \quad (\text{Bayes Rule})$$

Under Naive Bayes Assumption

$$P(x_1, \dots, x_n | y) = \prod_{i=1}^n P(x_i | y)$$

↳ Inference

$$P(y|x) = \frac{P(x|y) P(y)}{P(x)}$$

$$\prod_i P(x_i | y) P(y)$$

under Naive Bayes Assumption

$$\frac{\prod_i P(x_i | y) P(y)}{P(x)} = \frac{\sum_y \prod_i P(x_i | y) P(y)}{\sum_y \prod_i P(x_i | y) P(y)}$$

$$x_i | y = c \sim \text{Multinomial}(\theta_c | y = c)$$

$$\sum_{c=1}^C \theta_c | y = 1$$

$$x_i \in \{1, \dots, C\}$$

↓
ith value of x_i

$$\{x^i, y^i\}_{i=1}^m$$

$$L(\theta) = \log \left[\prod_{i=1}^m P(x^i, y^i; \Phi, \theta) \right]$$

$$\log \left[\prod_{i=1}^m \underbrace{P(x^i | y^i; \theta)}_{\downarrow} P(y^i; \Phi) \right]$$

Naive Bayes

Assumption:-

$$\prod_{j=1}^n P(x_j^i | y^i; \theta_j | y^i)$$